CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

Malaria is also regarded as a problem issue to the household earnings of the total areas in the world as the illness plagues other households in the areas. The decision makers in the field of the public health require prompt evidence based systems which would help the decision makers in the field to make informed decisions regarding the burden of the malaria at the house levels. The tools of prediction available does not reflect on the accuracy and cannot be utilized in the finished job of dealing with a complex data.

Another optimization of the latter version can be discussed as the alternative proposed in the given project in terms of the further prediction of the malaria-burden with the assistance of Structured Additive Regression algorithm. This independent test data was eventually validated so as to test this model that is trained and tested using house hold survey data of Mkpat Enin LGA, Akwa Ibom State.

The classical models were not quite realistic and comprehensible as compared to the next model. The ability of the public health departments to make sound decisions based on the implication of the malaria on the households, and predictions can significantly augment the capacity of the public health departments.

5.1 Conclusion

It is possible to utilize the given project and use Malaria-burden prediction model with Structured Additive Regression (STAR) model which assumes the application of household socio-economic, socio-demographic, cost and geospatial predictor in Mkpat Enin LGA, Akwa Ibom State. The use of STAR model was fairly effective. These findings show that a model of this nature is extremely consistent to respond

to high and low levels of malaria homes burden. The system provides semi-parametric modeling which can be easily interpreted to bring forth real time and evidence-based predicting on homes and fast in an easy web interface by using Streamlit. Those are the methods that have many flaws of the traditional statical models in their essence, and offer the practical assistance of the specific malaria prevention and the most applicable resources allocation.

5.2 Contributions to Knowledge

The issue of the overextended lower level that is pressing due to the level of the state and federal level may be regarded even nowadays, since the recent studies focus on the level of the household. The STAR model which has been adopted in the present paper splits the features in this work into additive spline features, which are easy to define nonlinear effect but the contribution of each of the features in this work is not understandable. The fact that Ridge and controlled spline complexity periodically normalize, also, causes the model to be normal that there were no constraints on the model, but instead it was free to assume any flexible relationships.

5.3 Recommendations

The medical staff must develop this predictive system so that it is not merely reacting to the medical interest of the Mkpat Enin and other LGA population so that it can be in a position to define the most appropriate burden households and communities when it comes to absorbing the interventions. An additional reflection required by the policymaker is the reflection of the system on the general surveillance systems to determine that the system output can be potent to national survey and the facility data. The integration will help in the micro-planning of the effect of the intervention and observing the levels in the house holds and the wards.

5.4 Suggestions for Future Work

a. It must expand sample to other states/LGAs other than Mkpat Enin LGA. Limit the question of the model efficiency and examine whether it can be applied to other conditions of ecology and socio-economic circumstances.

b. Test newer algorithms to the same data such as Bayesian hierarchical model, GAMs, gradient-boosting and deep learning models of STAR model. This will help in testing on paper trade off that is present between the interpretability and the computational efficiency and the accuracy.

c. investigate the accountability and the quality of the model as the errors that were made to the vulnerable categories of the population such as the poorest household as well as the women headed household. Broadened the paradigm/methodology of data collection in a way that could minimize the possible biases.